Supplementary Appendix

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This appendix has been provided by the authors to give readers additional information about the work.

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Supplementary Appendix

For

Event-free Survival with Pembrolizumab in Early Triple-Negative Breast Cancer

Authors: Schmid et al.

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List of Investigators.

Country	Site Name	Principal Investigator
Australia	Cabrini Health	Antill, Yoland
	Royal North Shore Hospital	Baron-Hay, Sally
	Westmead Hospital	Hui, Rina
	Royal Brisbane and Women's Hospital	Inglis, Po-ling
	Royal Adelaide Hospital	Murray, Nick
	Frankston Hospital	Thomson, Jacqui
	St. John of God Subiaco Hospital	Tsoi, Daphne
Brazil	Uniao Brasileira de Educacao e Assistencia Hospital Sao Lucas da Pucrs	Barrios, Carlos Henrique E.
	UOPECCAN - Uniao Oeste Paranaense de Estudos e Combate ao Cancer	da Cunha, Ademar Dantas Jr.
	Universidade de Caxias do Sul	de Cassia Costamilan, Rita
	Instituto do Cancer do Ceara	de Figueiredo Chaves, Fabio
	Hospital Araujo Jorge Associacao de Combate ao Cancer de Goias	de Freitas, Ruffo Jr.
	Hospital de Base de Sao Jose de Rio Preto	Girotto, Gustavo Colagiovanni
	Instituto do Cancer do Estado do Sao Paulo – ICESP	Linck, Rudinei Diogo Marques
	Hospital Nossa Senhora da Conceicao	Pedrini, Jose Luiz
	Hospital Erasto Gaertner	Skare, Nils Gunnar
Canada	Jewish General Hospital	Boileau, Jean-Francois
	Princess Margaret Cancer Centre	Cescon, David
	CIUSSS de l'Estrie-CHUS	Pavic, Michel
	CHU de Quebec Universite Laval - Hopital du Saint-Sacrement	Provencher, Louise/Poirier, Brigitte
	Centre Hospitalier de l Universite de Montreal – CHUM	Robidoux, Andre/Hassan, Saima
	The Ottawa Hospital - Cancer Care	Song, Xinni
	Tom Baker Cancer Centre	Verma, Sunil/Webster, Marc
Colombia	Hemato Oncologos S.A.	Avendano Rojas, Ana Cristina
	Instituto Nacional de Cancerologia (Bogota, Colombia)	Bruges, Ricardo
	Oncomedica S.A.	Gonzalez Fernandez, Manuel Enrique
	Instituto De Cancerologia S.A.	Montoya Restrepo, Maria Elvira/Gonzalez, Diego Mauricio
	Hospital Universitario San Ignacio	Rojas, Luis Leonardo/Sanchez Castillo, Jesus Oswaldo
	Oncologos del Occidente S.A.	Rojas-Uribe, Gustova Adolfo
France	Clinique Victor Hugo	Bourgeois, Hugues
	CHU de la Miletrie Poitiers	Chieze, Stephanie
	Hopital prive du Confluent	Chocteau-bouju, Dorothee
	Institut Claudius Regaud IUCT Oncopole	Dalenc, Florence

	Hopital Diaconesses Croix Saint Simon	Delbaldo, Catherine
	Polyclinique Bordeaux Nord Aquitaine	Dohollou, Nadine
	Hopital Saint Louis	Giacchetti, Sylvie
	Centre Francois Baclesse	Levy, Christelle
	Institut Curie	Loirat, Delphine/Sablin, Marie-Paule
	CHU Jean Minjoz	Mansi, Laura
	Centre Jean Perrin	Mouret-Reynier, Marie-Ange
Germany	Caritasklinik St. Theresia	Deryal, Mustafa
	Universitaetsklinikum Erlangen	Fasching, Peter A.
	Klinikum der Universit. Muenchen	Harbeck, Nadia
	Universitaets-Frauenklinik Tuebingen	Hartkopf, Andreas
	Gynaekologisches Zentrum	Kurbacher, Christian Martin
	Kliniken Essen Mitte	Reinisch, Mattea
	Universitaetsklinik und Poliklinik Halle/Saale	Thomssen, Christoph
	HELIOS Klinikum Berlin-Buch	Untch, Michael
	Universitaetsklinikum Hamburg-Eppendorf	Witzel, Isabell
Ireland	Bon Secours Hospital	Murphy, Conleth
	St Vincent's University Hospital	Walshe, Janice
Israel	Oncology Institute	Geffen, David/Tokar, Margarita
	Assaf Harofeh MC Zerifin	Katz, Daniella/Rasco, Adi
	Sheba Medical Center, Tel Hashomer Oncology Institute	Paluch-Shimon, Shani/Kaufman, Bella
	Hadassah Ein Karem – Sharett Institute of Oncology Kiryat Hadassah	Peretz-Yablonski, Tamar
	Tel Aviv Sourasky Medical Center	Ryvo, Larisa/Evron, Ella/Wolf, Ido
	Rabin-Medical Center	Stemmer, Salomon
Italy	Ospedale San Luca, Azienda Usl2 Toscana Nord Ovest	Baldini, Editta
	Ospedale Civile di Macerata	Beneditti, Giovanni/Battelli, Nicola
	Istituto Europeo di Oncologia	Colleoni, Marco
	Istituto Nazionale Tumori IRCCS Fondazione Pascale	De Laurentiis, Michelino
	Azienda Socio Sanitaria Territoriale degli Spedali Civili di Brescia	Pedersini, Rebecca
	Istituto Scientifico Romagnolo per Studio e Cura Tumori IRST	Rocco, Andrea/di Giorgi, Ugo
Japan	Aichi Cancer Center Hospital	Hattori, Masaya/Iwata, Hiroji
	St. Luke's International Hospital	Hayashi, Naoki
	Saitama Cancer Center	Inoue, Kenichi
	Kindai University Hospital	Komoike, Yoshifumi
	National Hospital Organization - Osaka National Hospital - Institute for Clinical Research	Masuda, Norikazu
	Hyogo College of Medicine Hospital	Miyoshi, Yasuo
	National Cancer Center Hospital East	Mukai, Hirofumi

	The Cancer Institute Hospital of JFCR	Ohno, Shinji
	Hiroshima City Hiroshima Citizens Hospital	Ohtani, Shoichiro/Itoh, Mitsuya
	Saitama Medical University International Medical Center	Saeki, Toshiaki/Osaki, Akihiko
	Social Medical Corporation Hakuaikai Sagara Hospital	Sagara, Yasuaki
	National Cancer Center Hospital	Tamura, Kenji/ Shimizu, Chikako/Yonemori, Kan
	National Hospital Organization Hokkaido Cancer Center	Takahashi, Masato
	Toranomon Hospital	Takano, Toshimi/Tanabe, Yuko
	Tokai University Hospital	Tokuda, Yutaka/Niikura, Naoki
	St. Marianna University School of Medicine Hospital	Tsugawa, Koichiro
	Shizuoka Cancer Center Hospital and Research Institute	Watanabe, Junichiro
	Chiba Cancer Center	Yamamoto, Naohito
	Kumamoto University Hospital	Yamamoto, Yutaka
Poland	Mazowiecki Szpital Onkologiczny	Bauer-Kosinska, Barbara
	Centrum Onkologii im. Prof. Franciszka Lukaszczyka	Chmielowska, Ewa
	Centrum Onkologii Ziemi Lubelskiej im. sw. Jana z Dukli	Chrzanowska-Kapica, Agata
	Narodowy Instytut Onkologii im. Marii Sklodowskiej-Curie - Oddzial w Krakowie	Grela-Wojewoda, Aleksandra/Drosik Kwasniewska, Anna
	Szpital Morski im. PCK. Szpitale Pomorskie Sp. Z o.o.	Kmak, Krzysztof Lesniewski/ Danielewicz, Iwona
	Dolnoslaskie Centrum Onkologii	Lacko, Aleksandra
	Narodowy Instytut Onkologii - Oddzial w Gliwicach	Tarnawski, Rafal
	Centrum Onkologii im. Prof. Franciszka Lukaszczyka	Zurawski, Bogdan
Portugal	Fundação Champalimaud	Cardoso, Fatima
	Instituto Português de Oncologia do Porto Francisco Gentil, EPE	Ferreira, Marta
	Hospital de Santa Maria, E.P.E.	Sousa, Rita Teixeira/ Dionísio, Maria Rita/Castro Abreu Ribeiro, Leonor
Republic of Korea	Asan Medical Center	Ahn, Jin Hee
	Seoul National University Hospital	Im, Seock-Ah
	Severance Hospital Yonsei University Health System	Kim, Gun Min/Kim, Min Hwan
	Samsung Medical Center	Park, Yeon Hee
Russia	Chelyabinsk Regional Clinical Oncological Dispensary	Fadeeva, Natalia Vladmirovna

	Scientific Research Oncology Institute n.a. N.N. Petrov	Krivorotko, Petr Vladimirovich
	Russian Oncological Research Center n.a. N.N. Blokhin of MoH	Lichinitser, Mikhail Romanovich/Ganshina, Inna
	Republican Clinical Oncology Dispensary of Republic of Bashkortostan	Lipatov, Oleg Nikolaevich
	Republican Clinical Oncology Dispensary of Tatarstan MoH	Mukhametshina, Guzel Zinnurovna
	Arkhangelsk Clinical Oncological Dispensary	Nechaeva, Marina N.
	GBU RO Regional Clinical Oncological Dispensary	Shomova, Marina V.
Singapore	National Cancer Centre Singapore	Dent, Rebecca
Spain	Hospital del Mar	Albanell Mestres, Joan
	Hospital Universitario Virgen del Rocio	Salvador Bofill, Francisco Javier
	Hospital Universitario Reina Sofia	de la Haba Rodriguez, Juan
	Hospital Clinico Univ de Valencia	Bermejo de las Heras, Begona
	Hospital Quiron de Madrid	Gonzalez Cortijo, Lucia
	Hospital Universitario Ramon y Cajal	Holgado, Esther/Gion, Maria
	Complejo Hospitalario Universitario de Santiago	Lopez Lopez, Rafael
	ICO L Hospitalet	Morilla Ruiz, Idoia /Villanueva, Rafael
	Hospital General Universitari Vall d Hebron	Munoz Couselo, Eva/Zamora Adelantado, Esther
	Instituto Oncologico Baselga, Hospital Quiron	Perez Garcia, Jose Manuel/Soberino, Jesus
Sweden	Norrlands Universitetssjukhus Universitetssjukhuset Cancercentrum	Andersson, Anne
	Karolinska Universitetssjukhuset Solna	Foukakis, Theodoros
	Linkopings Universitetssjukhus Onkologiska kliniken	Hallbeck, Anna-Lotta
	Akademiska Sjukhuset Onkologiska kliniken	Lindman, Henrik
Taiwan	MacKay Memorial Hospital	Chang, Yuan-Ching
	Chang Gung Medical Foundation, Linkou Branch	Chen, Shin-Cheh
	National Taiwan University Hospital	Huang, Chiun-Sheng
	National Cheng Kung University Hospital	Lee, Kuo-Ting
	Koo Foundation Sun Yat-Sen Cancer Center	Liu, Mei-Ching/Chung, Chi-Feng
	Taipei Veterans General Hospital	Tseng, Ling-Ming
Turkey	Adana Acıbadem Hospital	Abali, Huseyin
	Hacettepe Universitesi Tip Fakultesi Hastanesi	Aksoy, Sercan
	Izmir Medical Park Hospital	Arslan, Cagatay
	Acibadem Altunizade Hastanesi	Basaran, Gul

	Trakya University Medical Faculty Balkan Oncology Hospital	Cicin, Irfan
	Memorial Sisli Hastanesi	Keskin, Serkan
	Amerikan Hospital Medical	Mandel, Nil Molinas
	Dr Abdurrahman Yurtaslan Oncology Training and Research Hospital	Oksuzoglu, Berna
	Ozel Medicana International Ankara Hastanesi	Ozdemir, Evren
	Antalya Memorial Hospital	Ozdogan, Mustafa
	İstanbul University Cerrahpaşa Medical Faculty	Ozguroglu, Mustafa
	Baskent Unıversity Adana Kısla Hospital	Ozyilkan, Ozgur
	Ege University Medical Faculty Tulay Aktas Oncology Hospital	Uslu, Ruchan F./Gokmen, Erhan
	Samsun Ondokuz Mayis Universitesi Tip Fakultesi Hastanesi Tip Fakultesi	Yucel, Idris/Yilmaz, Bahiddin
United Kingdom	Nottingham University Hospitals NHS Trust	Chan, Steve
	The James Cook University Hospital	Graham, Janine Elizabeth
	Maidstone Hospital	Harper-Wynne, Catherine
	St George's Hospital	Kyle, Fiona
	Colchester General Hospital	Mukesh, MB
	Barts Cancer Institute, Charterhouse Square, Queen Mary University of London	Schmid, Peter
	Royal Cornwall Hospitals NHS Trust	Wheatley, Duncan
United States	Henry Ford Hospital	Ali, Haythem
	Compass Oncology	Andersen, Jay C.
	New England Cancer Specialists	Battelli, Chiara
	Goshen Center for Center Care	Bruetman, Daniel G./Kio, Ebenezer
	Houston Methodist Cancer Center	Chang, Jenny
	Medical Oncology Associates (Summit Cancer Centers)	Chaudhry, Arvind
	Virginia Oncology Associates	Danso, Michael A.
	Virginia Cancer Specialists, PC	Denduluri, Neelima/Spira, Alexander I.
	University of Colorado Cancer Center	Diamond, Jennifer Robinson
	University of Virginia; Div. Hematology-Oncology	Dillon, Patrick
	Helen F. Graham Cancer Center & Research Institute	Guarino, Michael J.
	Moncrief Cancer Institute	Haley, Barbara B.
	Parkland Health and Hospital System	Haley, Barbara B.
	Simmons Cancer Center	Haley, Barbara B.
	UT Southwestern Medical Center	Haley, Barbara B.
	Broome Oncology, LLC	Harris, Ronald P.
	Texas Oncology-Memorial City	Holmes, Frankie Ann/Cairo, Michelina
	Bon Secours Cancer Institute	Irvin, William J.

	Magee Women's Hospital	Jankowitz, Rachel/Emens, Leisha Ann
	YVMH dba Virginia Mason Memorial/North Star Lodge Cancer Center	Jones, Vicky E./Obulareddy, Sri
	University of Iowa Hospital and Clinics	Leone, Jose/Phadke, Sneha
	Rhode Island Hospital	Lopresti, Mary
	Peninsula Cancer Institute, LLC	MacLaughlin, William
	Cedars Sinai Medical Center	McArthur, Heather
	TriHealth Cancer Institute-Good Samaritan Hospital	Mehta, Apurva
	University of Chicago Medical Center	Nanda, Rita
	Orchard Healthcare Research Inc.	Oliff, Ira A.
	Texas Oncology - Baylor Charles A. Sammons Cancer Center	OShaughnessy, Joyce A.
	Providence Portland Medical Center	Page, David
	Texas Oncology-Austin Central	Patt, Debra A.
	Yale University School of Medicine	Pusztai, Lajos
	Texas Oncology-Tyler	Richards, Donald A.
	Nyack Hospital Infusion Center	Rybalova, Irina
	Scottsdale Healthcare Hospitals D/B/A HonorHealth	Sachdev, Jasgit/Gordon, Michael S.
	NorthShore University Health System	Saha, Poornima
	Seattle Cancer Care Alliance	Specht, Jennifer
	Pacific Cancer Care	Stampleman, Laura/Koontz, Michael Zachary
	Texas Oncology-Plano East	Stokoe, Christopher T.F.
	Arizona Oncology Associates P.C.- HOPE	Swart, Rachel
	Innovative Clinical Research Institute	Thara, Eddie
	Rutgers Cancer Institute of New Jersey	Toppmeyer, Deborah Lynn
	Minnesota Oncology Hematology, P.A.	Tsai, Michaela
	University of Miami - Sylvester Comprehensive Cancer Center - Kendall satellite	Valdes-Albini, Frances
	The West Clinic, P.C.	Vidal, Gregory
	Oncology Hematology Care, Inc.	Ward, Patrick
	North Shore Oncology Hematology Associates	Weyburn, Thomas D./Chung, Elizabeth K.
	Texas Oncology-San Antonio Northeast	Wilks, Sharon T./Harroff, Allyson
	Kadlec Clinic Hematology and Oncology	Zhuo, Ying

Steering Committee.

Peter Schmid: Barts Cancer Institute, Queen Mary University London, London, UK

Javier Cortes: International Breast Cancer Center, Quironsalud Group; Vall d'Hebron Institute of Oncology (VHIO), Madrid & Barcelona, Spain and Universidad Europea de Madrid, Faculty of Biomedical and Health Sciences, Department of Medicine, Madrid, Spain

Carsten Denkert: Institute of Pathology, Philipps-University Marburg and University Hospital Marburg (UKGM), Marburg, Germany

Rebecca Dent: National Cancer Center Singapore, Duke-NUS Medical School, Singapore

Sherko Kuemmel: Breast Unit, Kliniken Essen-Mitte, Essen, Germany and Charité – Universitätsmedizin Berlin, Department of Gynecology with Breast Center, Berlin, Germany

Heather McArthur: University of Texas Southwestern Medical Center, Dallas, TX, USA

Lajos Pusztai: Yale School of Medicine, Yale Cancer Center, New Haven, CT, USA

Data Monitoring Committee.

Massimo Cristofanilli: Northwestern University, Feinberg School of Medicine, Chicago, IL, USA

James Dignam: The University of Chicago, Chicago, IL, USA

Ian Krop: Dana-Farber/Harvard Cancer Center, Boston, MA, USA

Kathy Miller: Indiana University Melvin and Bren Simon Cancer Center, Indianapolis, IN, USA

Supplemental Statistical Methods.

Adjustment for multiplicity

The following pre-specified multiplicity strategy was applied to the two primary hypotheses and the secondary hypothesis. The two primary hypotheses test superiority of pembrolizumab compared to placebo in pCR (ypT0/Tis ypN0) or EFS in patients with early TNBC. The secondary hypothesis tests superiority in OS. The overall Type-I error among multiple end points is strongly controlled at 5% (two-sided), with 1% initially allocated to the pCR (ypT0/Tis ypN0) hypothesis and 4% initially allocated to the EFS hypothesis. The study is considered successful if pCR (ypT0/Tis ypN0) or EFS is demonstrated to be statistically significant at either an interim analysis or the final analysis under multiplicity control.

The study used the graphical method of Maurer and Bretz to control multiplicity for multiple hypotheses and interim analyses. According to this approach, the study hypotheses may be tested more than once, and when a particular null hypothesis is rejected, the α allocated to that hypothesis can be reallocated to other hypothesis tests.

Supplemental Methods.

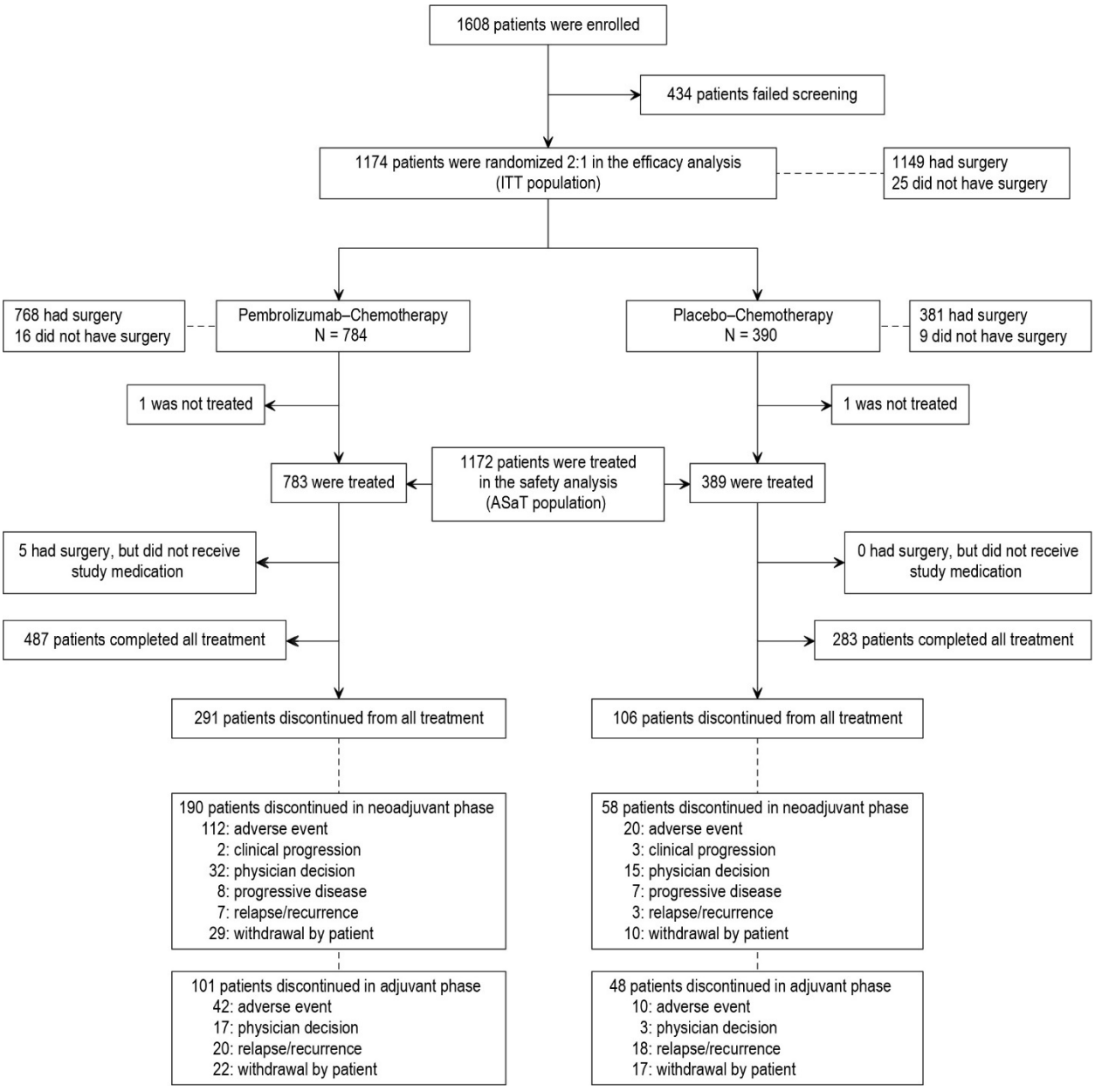
Prespecified subgroups

To evaluate the consistency of the treatment effects across various subgroups, the estimate of the between-group treatment effect (with a nominal 95% CI) for the primary endpoints was estimated and plotted within each category of the following classification variables:

- Nodal status: Positive vs. Negative
- Tumor size: T1/T2 vs. T3/T4
- Choice of Carboplatin: Q3W vs. Weekly
- Tumor PD-L1 status: positive (CPS ≥1) vs. negative (CPS <1)
- Menopausal status: pre-menopausal vs. post-menopausal
- Age: <65 years vs. ≥65 years
- Geographic region: Europe/Israel/North America/Australia vs. Asia vs. Rest of World
- Ethnic origin: Hispanic vs. Non-Hispanic
- ECOG performance status: 0 vs. 1
- HER2 status: IHC 2+ (but FISH-) vs. IHC 0-1+
- LDH: >upper limit of normal (ULN) vs. ≤ULN

Figure S1. Trial Enrollment and Outcomes of the Patients at the Fourth Interim Analysis.

Shown are data until the data-cutoff date of March 23, 2021. Patients did not have to complete all neoadjuvant therapy to undergo surgery. ITT, intention-to-treat.



No patients remained on treatment at the fourth interim analysis.

Fig. S2. Additional Subgroup Analyses for Event-Free Survival. An analysis of event-free survival in additional subgroups is shown. For the overall population, the analysis was based on the Miettinen and Nurminen method stratified according to nodal status (positive or negative), tumor size (T1 to T2 or T3 to T4), and frequency of carboplatin administration (once weekly or once every 3 weeks). For the other subgroups, the analysis was based on the unstratified Miettinen and Nurminen method. Patients with missing values in a subgroup category variable are not included in the subgroup analysis.

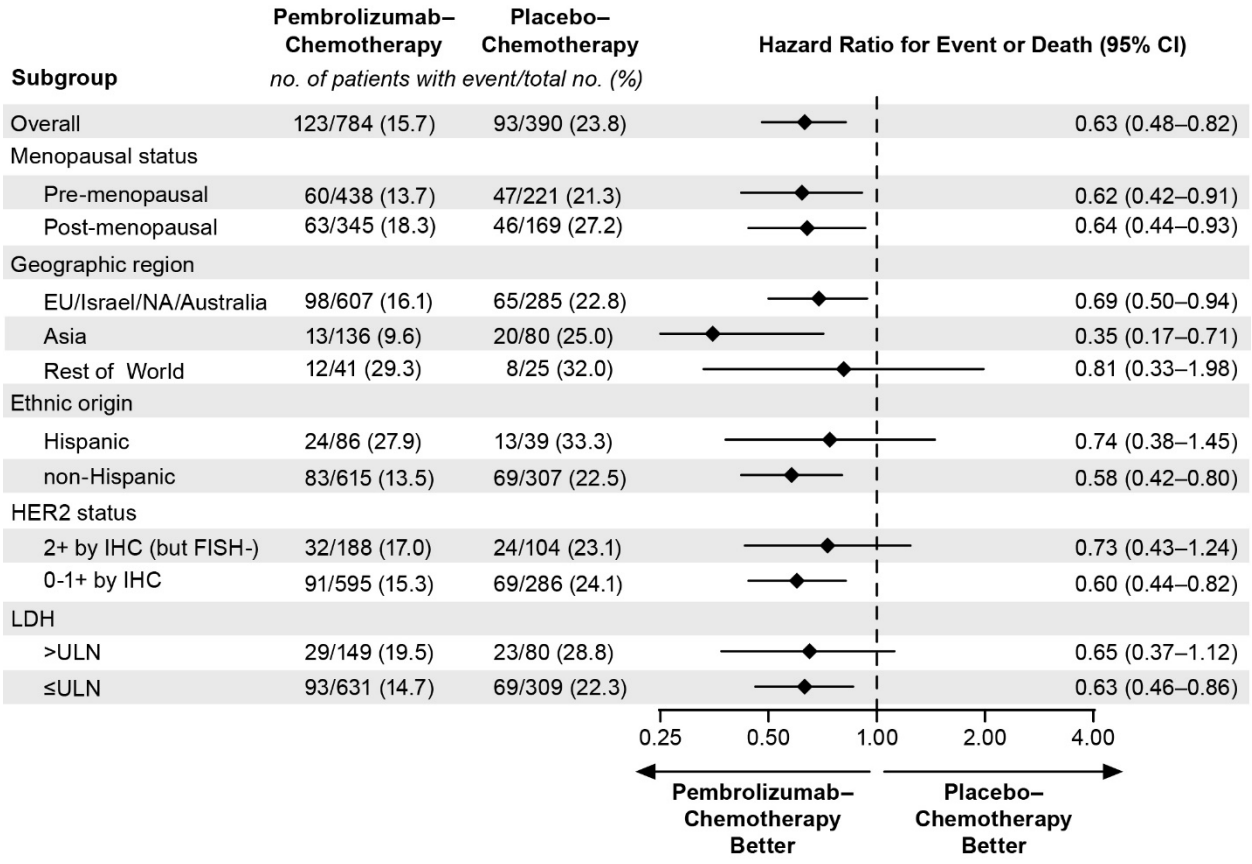
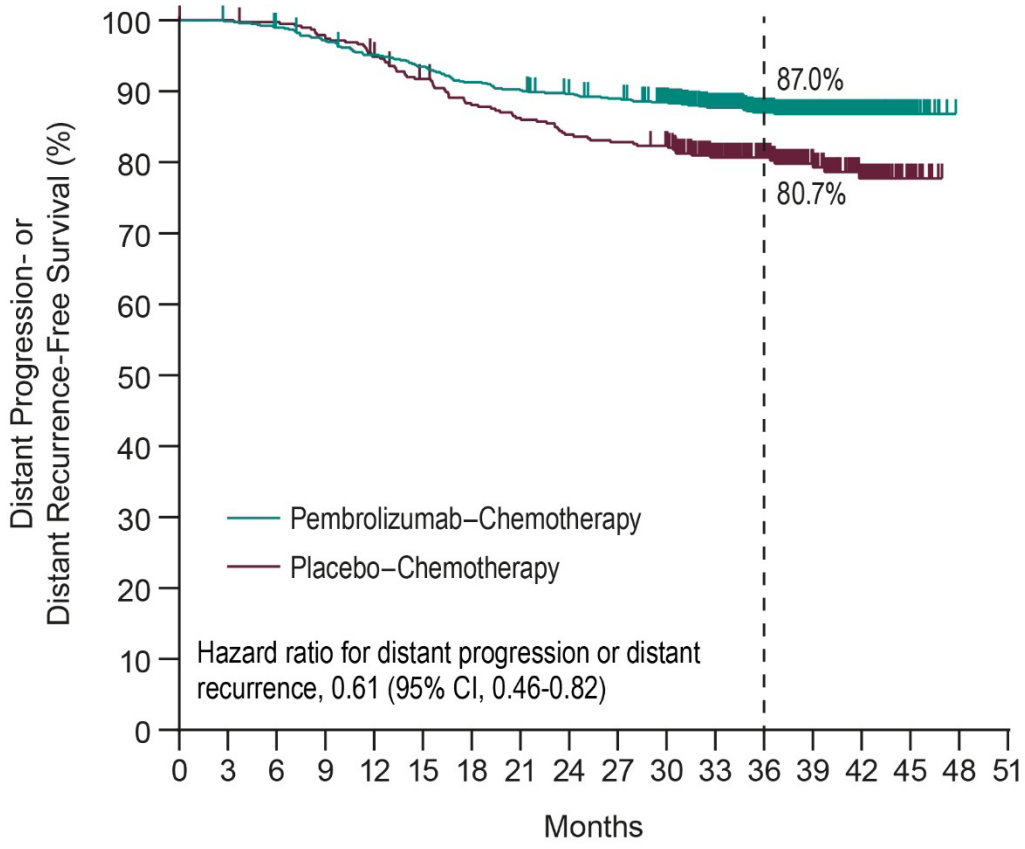


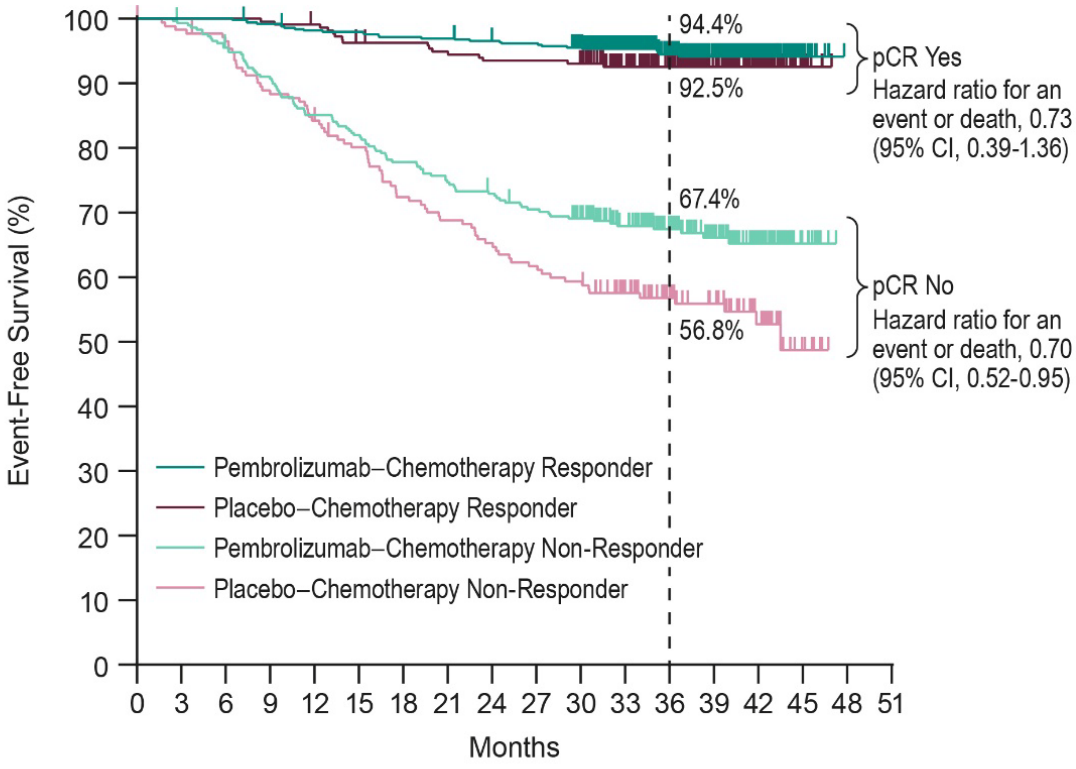
Fig. S3. Kaplan-Meier Estimates of Distant Progression- or Distant Recurrence- Free Survival According to Treatment Group in the Intention-to-Treat Population. Tick marks represent data censored at the last time the patient was known to be alive without an event (distant progression; distant recurrence; or death from any cause) for patients who did not experience an event. The hazard ratio and CI were analyzed based on a Cox regression model with treatment as a covariate stratified by the randomization stratification factors of nodal status (positive versus negative), tumor size (T1 to T2 or T3 to T4), and frequency of carboplatin administration (once weekly or once every 3 weeks).



No. at Risk

Pembrolizumab–Chemotherapy	784	782	773	758	741	728	711	702	692	685	663	561	439	308	167	29	0	0
Placebo–Chemotherapy	390	389	387	379	367	352	337	330	321	317	312	259	202	143	84	17	0	0

Fig. S4. Kaplan-Meier Estimates of Event-Free Survival by Pathological Complete Response (ypT0/Tis ypN0) According to Treatment Group in the Intention-to-Treat Population. Tick marks represent data censored at the last time the patient was known to be alive and without an event (disease progression that precludes definitive surgery; local or distant recurrence or a second primary tumor; or death from any cause) for patients who did not experience an event. The hazard ratio and CI were analyzed based on a Cox regression model with treatment as a covariate.



No. at Risk

Pembrolizumab–Chemotherapy Responder	494	494	494	489	483	482	478	477	472	470	460	387	307	220	122	18	0	0
Placebo–Chemotherapy Responder	217	217	217	216	214	207	206	203	200	200	197	165	130	87	56	9	0	0
Pembrolizumab–Chemotherapy Non-Responder	290	287	275	262	245	236	224	215	209	201	192	164	126	83	43	10	0	0
Placebo–Chemotherapy Non-Responder	173	169	165	152	144	135	122	116	110	104	100	85	65	53	27	8	0	0

Table S1. Demographic and Disease Characteristics at Baseline.*

Characteristic	Pembrolizumab–Chemotherapy (N=784)	Placebo–Chemotherapy (N=390)
Age		
Median (range) — yr	49 (22-80)	48 (24-79)
<65 yr — no. (%)	700 (89.3)	342 (87.7)
Female sex — no. (%)	783 (99.9)	390 (100.0)
Geographic region — no. (%)		
Europe	388 (49.5)	180 (46.2)
Asia	166 (21.2)	91 (23.3)
North America	166 (21.2)	78 (20.0)
Australia	23 (2.9)	16 (4.1)
Rest of world	41 (5.2)	25 (6.4)
Race		
White	504 (64.3)	242 (62.1)
Asian	149 (19.0)	89 (22.8)
Missing†	65 (8.3)	31 (7.9)
Black or African American	38 (4.8)	15 (3.8)
American Indian or Alaska Native	14 (1.8)	7 (1.8)
Multiple	13 (1.7)	6 (1.5)
Native Hawaiian or Other Pacific Islander	1 (0.1)	0
Ethnicity		
Not Hispanic or Latino	615 (78.4)	307 (78.7)
Hispanic or Latino	86 (11.0)	39 (10.0)
Not Reported	46 (5.9)	28 (7.2)
Unknown	19 (2.4)	11 (2.8)
Missing†	18 (2.3)	5 (1.3)
Menopausal status — no. (%)		
Pre-menopausal	438 (55.9)	221 (56.7)
Post-menopausal	345 (44.0)	169 (43.3)
Missing‡	1 (0.1)	0 (0.0)
PD-L1 status§ — no. (%)		
Positive	656 (83.7)	317 (81.3)
Negative	128 (16.3)	69 (17.7)
Missing	0 (0.0)	4 (1.0)
ECOG performance status		
0	678 (86.5)	341 (87.4)
1	106 (13.5)	49 (12.6)
Lactase dehydrogenase — no. (%)		
≤ULN	631 (80.5)	309 (79.2)
>ULN	149 (19.0)	80 (20.5)
Missing	4 (0.5)	1 (0.3)
Choice of carboplatin‖ — no. (%)		
Every 3 weeks	335 (42.7)	167 (42.8)
Weekly	449 (57.3)	223 (57.2)
Primary tumor classification‖— no. (%)		
T1/T2	580 (74.0)	290 (74.4)
T3/T4	204 (26.0)	100 (25.6)
Nodal involvement‖ — no. (%)		
Positive	405 (51.7)	200 (51.3)
Negative	379 (48.3)	190 (48.7)

Overall disease stage — no. (%)		
I	0 (0.0)	1 (0.3)
II	590 (75.3)	291 (74.6)
III	194 (24.7)	98 (25.1)
HER2 status — no. (%)		
0-1 by IHC	595 (75.9)	286 (73.3)
2+ by IHC¶	188 (24.0)	104 (26.7)
Missing	1 (0.1)	0 (0.0)

*Based on data from the intention-to-treat population. Percentages may not total 100 because of rounding. PD-L1 denotes programmed death ligand 1; ECOG denotes Eastern Cooperative Oncology Group; HER2 denotes human epidermal growth factor receptor 2; IHC denotes immunohistochemistry.
†Information on race and ethnicity of trial participants is based on self-report and unavailable in some countries due to local laws and practices.
‡From one male patient.
§The PD-L1 combined positive score was defined as number of PD-L1–positive cells (tumor cells, lymphocytes, and macrophages) divided by total number of tumor cells × 100. PD-L1 positivity was defined as CPS ≥1.
‖Categories of these classification variables based on the randomization stratification factors are from a central interactive voice-response system with an integrated Web-response system.
¶All tumors with HER2 expression 2+ by IHC were negative for HER2 amplification by in situ hybridization.

Table S2. Representativeness of Study Participants.

Disease under investigation	Triple-negative breast cancer.
Special considerations related to	
Sex and gender	Breast cancer is the most common malignancy among women worldwide[1], with triple-negative breast cancer accounting for 15% to 20% of all breast cancer subtypes.[2] Triple-negative breast cancer in men is extremely rare.
Age	Compared with other forms of breast cancer, relative rates of triple-negative breast cancer are higher in young persons, with many patients being diagnosed before the age of 50 years.[3,4]
Race or ethnic group	Triple-negative breast cancer affects Black persons disproportionately in the United States.[5] Several studies suggest a similar rate of triple-negative breast cancer among Hispanic and non-Hispanic White persons.[6] Compared with non-Hispanic White persons, Asian persons have significantly lower rates of premenopausal triple-negative breast cancer, but rates do not decline with increasing age.[7]
Geography	In the 2018 Global Cancer Observatory report, triple-negative breast cancer mortality rates ranged from 15.4 to 18.4 per 100,000 in Africa; rates were lower in Europe and North America despite incidences double to triple that observed in Africa. There is significant heterogeneity among different groups of Black persons, with the highest rates observed in West African (24.1% of cases) and African American women (23.7%), and lower rates observed in Caribbean (21.2%) and East African (11.6%) women.[8] There is also significant heterogeneity within the Hispanic population, with rates varying among Cuban, Mexican, and Puerto Rican persons.[9]
Other considerations	Throughout the world, incidence and mortality rates vary greatly within and between countries. Ethnic predisposition, family history, genetic associations and modifiable risk factors are inextricably intertwined.
Overall representativeness of this trial	The participants in this trial demonstrated the expected sex and age distribution. The proportion of Black persons who underwent randomization was small (4.5%). Among participants enrolled in North America, 15.2% were Black, a relatively larger proportion than the total population distribution of Black persons in the United States. Most participants enrolled were in Europe (48.4%) and North America (20.8%); no patient was enrolled in Africa. Information on race and ethnicity of trial participants is based on self-report and unavailable in some countries due to local laws and practices.

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Table S3. Summary of Drug Exposure in the Combined Phases.

Event	Pembrolizumab–Chemotherapy (N=783)	Placebo–Chemotherapy (N=389)
All drugs		
n	778	389
Median weeks (range)	57.9 (0.1 – 95.3)	59.1 (0.1 – 86.1)
Pembrolizumab (200 mg Q3W)		
n	778	0
Median weeks (range)	57.9 (0.1 – 95.3)	0
Median no. administrations (range)	17.0 (1.0. – 17.0)	0
Placebo (Q3W)		
n	0	389
Median weeks (range)	0	59.1 (0.1 – 86.1)
Median no. administrations (range)	0	17.0 (1.0 – 17.0)
Carboplatin (weekly)		
n	444	220
Median weeks (range)	11.1 (2.0 – 26.1)	11.1 (2.3 – 17.1)
Median no administrations (range)	12.0 (3.0 – 12.0)	12.0 (3.0 – 12.0)
Carboplatin (Q3W)		
n	334	167
Median weeks (range)	9.6 (0.1 – 16.4)	9.4 (3.0 – 15.1)
Median no administrations (range)	4.0 (1.0 – 4.0)	4.0 (2.0 – 4.0)
Paclitaxel (weekly)		
n	778	389
Median weeks (range)	11.3 (0.1 – 26.1)	11.3 (0.1 – 17.1)
Median no. administrations (range)	12.0 (1.0 – 13.0)	12.0 (1.0 – 12.0)
Doxorubican (Q3W)		
n	488	247
Median weeks (range)	9.1 (0.1 – 13.1)	9.1 (0.1 – 13.3)
Median no. administrations (range)	4.0 (1.0 – 4.0)	4.0 (1.0 – 4.0)
Epirubicin (Q3W)		
n	238	122
Median weeks (range)	9.1 (0.1 – 13.1)	9.1 (0.1 – 12.0)
Median no administrations (range)	4.0 (1.0 – 5.0)	4.0 (1.0 – 4.0)
Cyclophosphamide (Q3W)		
n	726	369
Median weeks (range)	9.1 (0.1 – 13.1)	9.1 (0.1 – 13.3)
Median no administrations (range)	4.0 (1.0 – 4.0)	4.0 (1.0 – 4.0)

Table S4. Adverse Events during the Adjuvant Phase.

Event	Pembrolizumab– Chemotherapy (N=588)		Placebo–Chemotherapy (N=331)	
	Any Grade	Grade ≥3	Any Grade	Grade ≥3
	number of patients (percent)			
Any adverse event*	542 (92.2)	88 (15.0)	294 (88.8)	38 (11.5)
Treatment-related adverse event†	316 (53.7)	37 (6.3)	161 (48.6)	9 (2.7)
Immune-mediated adverse event‡	50 (8.5)	17 (2.9)	16 (4.8)	1 (0.3)
Hypothyroidism	17 (2.9)	1 (0.2)	12 (3.6)	0
Severe skin reaction	11 (1.9)	6 (1.0)	0	0
Pneumonitis	7 (1.2)	4 (0.7)	2 (0.6)	1 (0.3)
Hyperthyroidism	5 (0.9)	0	2 (0.6)	0
Adrenal insufficiency	3 (0.5)	0	0	0
Colitis	2 (0.3)	1 (0.2)	0	0
Myocarditis	2 (0.3)	1 (0.2)	0	0

*Listed are all adverse events that occurred during adjuvant treatment period or within the 30 days thereafter (within 90 days for serious events). The as-treated population included all patients who underwent randomization and received one adjuvant treatment. Events are listed in descending order of frequency in the pembrolizumab-chemotherapy group. The severity of adverse events was graded according to the National Cancer Institute Common Terminology Criteria for Adverse Events, version 4.0.

†Adverse events that were attributed to study treatment by the investigator. There were no treatment-related adverse events that occurred in at least 15% of patients. Grade 5 treatment-related adverse events were pulmonary embolism (n=1) and autoimmune encephalitis (n=1) in the pembrolizumab–chemotherapy group.

‡Adverse events based on a list of terms specified by the sponsor and considered regardless of treatment attribution by the investigator that occurred in at least 2 patients are reported. One grade 5 immune-mediated adverse event of autoimmune encephalitis occurred in the pembrolizumab–chemotherapy group.

Table S5. Summary of Tier 2 Adverse Events during the Combined Phases.

Participants in Population	Pembrolizumab–Chemotherapy (N=783)		Placebo–Chemotherapy(N=389)		Difference in % vs. Placebo–Chemotherapy
	n	(%)	n	(%)	Estimate (95% CI)*
with one or more adverse events	777	(99.2)	389	(100.0)	-0.8 (-1.7, 0.2)
with no adverse event	6	(0.8)	0	(0.0)	0.8 (-0.2, 1.7)
with drug-related† adverse events	774	(98.9)	388	(99.7)	-0.9 (-2.0, 0.4)
with toxicity grade 3-5‡ adverse events	645	(82.4)	306	(78.7)	3.7 (-1.0, 8.7)
with toxicity grade 3-5‡ drug-related adverse events	604	(77.1)	285	(73.3)	3.9 (-1.3, 9.3)
with serious adverse events	341	(43.6)	111	(28.5)	15.0 (9.2, 20.6)
with serious drug-related adverse events	267	(34.1)	78	(20.1)	14.0 (8.7, 19.1)
with dose modification§ due to an adverse event	644	(82.2)	306	(78.7)	3.6 (-1.1, 8.6)
who died	7	(0.9)	1	(0.3)	0.6 (-0.6, 1.6)
who died due to a drug-related adverse event	4	(0.5)	1	(0.3)	0.3 (-1.0, 1.1)
discontinued drug due to an adverse event	234	(29.9)	60	(15.4)	14.5 (9.5, 19.1)
discontinued drug due to a drug-related adverse event	217	(27.7)	55	(14.1)	13.6 (8.7, 18.1)
discontinued drug due to a serious adverse event	94	(12.0)	15	(3.9)	8.1 (5.0, 11.1)
discontinued drug due to a serious drug-related adverse event	84	(10.7)	11	(2.8)	7.9 (5.0, 10.6)

Includes all adverse events started from the first treatment including definitive surgery and radiation therapy and up to 30 days of the last treatment including definitive surgery and radiation therapy for the non-serious adverse events and up to 90 days of the last treatment including definitive surgery and radiation therapy for the serious adverse events.

*Based on Miettinen & Nurminen method; since 95% CIs are provided without adjustment for multiplicity, they should be regarded as a helpful descriptive measure, not a formal method for assessing the statistical significance of the between-group differences in adverse experiences.

†Determined by the investigator to be related to the drug.

‡Grades are based on NCI CTCAE version 4.0.

§Defined as an action taken of dose reduced, drug interrupted, or drug withdrawn.

Table S6. Adverse Events Occurring at an Incidence ≥5% in One or More Treatment Groups in the Combined Phases.

Participants in Population	Pembrolizumab–Chemotherapy (N=783)		Placebo–Chemotherapy (N=389)		Difference in % vs. Placebo–Chemotherapy
	n	(%)	n	(%)	Estimate (95% CI)*
with one or more adverse events	777	(99.2)	389	(100.0)	-0.8 (-1.7, 0.2)
with no adverse events	6	(0.8)	0	(0.0)	0.8 (-0.2, 1.7)
Blood and lymphatic system disorders	**608**	**(77.7)**	**314**	**(80.7)**	**-3.1 (-7.8, 2.0)**
Anemia	463	(59.1)	229	(58.9)	0.3 (-5.7, 6.3)
Febrile neutropenia	151	(19.3)	66	(17.0)	2.3 (-2.5, 6.8)
Leukopenia	98	(12.5)	51	(13.1)	-0.6 (-4.9, 3.3)
Neutropenia	376	(48.0)	190	(48.8)	-0.8 (-6.9, 5.2)
Thrombocytopenia	110	(14.0)	68	(17.5)	-3.4 (-8.1, 0.9)
Cardiac disorders	**114**	**(14.6)**	**48**	**(12.3)**	**2.2 (-2.1, 6.2)**
Palpitations	32	(4.1)	21	(5.4)	-1.3 (-4.3, 1.1)
Ear and labyrinth disorders	**74**	**(9.5)**	**32**	**(8.2)**	**1.2 (-2.4, 4.5)**
Endocrine disorders	**184**	**(23.5)**	**33**	**(8.5)**	**15.0 (10.8, 19.0)**
Hyperthyroidism	41	(5.2)	7	(1.8)	3.4 (1.2, 5.5)
Hypothyroidism	118	(15.1)	22	(5.7)	9.4 (5.9, 12.8)
Eye disorders	**157**	**(20.1)**	**64**	**(16.5)**	**3.6 (-1.2, 8.1)**
Dry eye	52	(6.6)	20	(5.1)	1.5 (-1.6, 4.2)
Gastrointestinal disorders	**680**	**(86.8)**	**351**	**(90.2)**	**-3.4 (-7.0, 0.6)**
Abdominal pain	112	(14.3)	49	(12.6)	1.7 (-2.6, 5.7)
Abdominal pain upper	80	(10.2)	34	(8.7)	1.5 (-2.3, 4.9)
Constipation	328	(41.9)	150	(38.6)	3.3 (-2.7, 9.2)
Diarrhea	318	(40.6)	133	(34.2)	6.4 (0.5, 12.2)
Dry mouth	66	(8.4)	30	(7.7)	0.7 (-2.8, 3.9)
Dyspepsia	111	(14.2)	56	(14.4)	-0.2 (-4.7, 3.9)
Gastroesophageal reflux disease	57	(7.3)	43	(11.1)	-3.8 (-7.6, -0.4)
Hemorrhoids	52	(6.6)	17	(4.4)	2.3 (-0.7, 4.9)
Nausea	522	(66.7)	257	(66.1)	0.6 (-5.1, 6.4)

Stomatitis	141	(18.0)	58	(14.9)	3.1 (-1.5, 7.4)
Vomiting	244	(31.2)	108	(27.8)	3.4 (-2.2, 8.8)
General disorders and administration site conditions	**668**	**(85.3)**	**325**	**(83.5)**	**1.8 (-2.5, 6.4)**
Asthenia	219	(28.0)	111	(28.5)	-0.6 (-6.1, 4.8)
Chills	42	(5.4)	8	(2.1)	3.3 (1.0, 5.4)
Fatigue	365	(46.6)	168	(43.2)	3.4 (-2.6, 9.4)
Influenza like illness	54	(6.9)	26	(6.7)	0.2 (-3.1, 3.1)
Mucosal inflammation	112	(14.3)	49	(12.6)	1.7 (-2.6, 5.7)
Oedema peripheral	70	(8.9)	30	(7.7)	1.2 (-2.3, 4.4)
Pyrexia	221	(28.2)	72	(18.5)	9.7 (4.6, 14.6)
Hepatobiliary disorders	**43**	**(5.5)**	**23**	**(5.9)**	**-0.4 (-3.5, 2.3)**
Immune system disorders	**78**	**(10.0)**	**30**	**(7.7)**	**2.2 (-1.4, 5.5)**
Hypersensitivity	40	(5.1)	10	(2.6)	2.5 (0.1, 4.7)
Infections and infestations	**525**	**(67.0)**	**265**	**(68.1)**	**-1.1 (-6.7, 4.7)**
Nasopharyngitis	65	(8.3)	52	(13.4)	-5.1 (-9.2, -1.4)
Upper respiratory tract infection	106	(13.5)	47	(12.1)	1.5 (-2.8, 5.3)
Urinary tract infection	123	(15.7)	62	(15.9)	-0.2 (-4.9, 4.1)
Injury, poisoning and procedural complications	**297**	**(37.9)**	**169**	**(43.4)**	**-5.5 (-11.5, 0.4)**
Infusion related reaction	79	(10.1)	27	(6.9)	3.1 (-0.4, 6.3)
Procedural pain	58	(7.4)	26	(6.7)	0.7 (-2.6, 3.7)
Radiation skin injury	114	(14.6)	73	(18.8)	-4.2 (-9.0, 0.2)
Investigations	**515**	**(65.8)**	**271**	**(69.7)**	**-3.9 (-9.4, 1.9)**
Alanine aminotransferase increased	238	(30.4)	108	(27.8)	2.6 (-3.0, 8.0)
Aspartate aminotransferase increased	187	(23.9)	77	(19.8)	4.1 (-1.0, 8.9)
Blood alkaline phosphatase increased	45	(5.7)	25	(6.4)	-0.7 (-3.9, 2.1)
Blood lactate dehydrogenase increased	36	(4.6)	20	(5.1)	-0.5 (-3.5, 1.9)
Neutrophil count decreased	191	(24.4)	113	(29.0)	-4.7 (-10.2, 0.7)
Platelet count decreased	78	(10.0)	37	(9.5)	0.5 (-3.4, 3.9)
Weight decreased	57	(7.3)	16	(4.1)	3.2 (0.3, 5.8)
White blood cell count decreased	113	(14.4)	56	(14.4)	0.0 (-4.4, 4.2)
Metabolism and nutrition disorders	**347**	**(44.3)**	**149**	**(38.3)**	**6.0 (0.0, 11.9)**

Decreased appetite	178	(22.7)	65	(16.7)	6.0 (1.1, 10.6)
Dehydration	39	(5.0)	9	(2.3)	2.7 (0.3, 4.8)
Hyperglycemia	36	(4.6)	23	(5.9)	-1.3 (-4.4, 1.3)
Hypokalemia	88	(11.2)	24	(6.2)	5.1 (1.6, 8.2)
Hypomagnesaemia	44	(5.6)	17	(4.4)	1.2 (-1.6, 3.7)
Musculoskeletal and connective tissue disorders	**479**	**(61.2)**	**249**	**(64.0)**	**-2.8 (-8.6, 3.1)**
Arthralgia	225	(28.7)	120	(30.8)	-2.1 (-7.8, 3.4)
Back pain	97	(12.4)	63	(16.2)	-3.8 (-8.3, 0.4)
Bone pain	70	(8.9)	39	(10.0)	-1.1 (-4.9, 2.3)
Myalgia	153	(19.5)	73	(18.8)	0.8 (-4.2, 5.4)
Neck pain	20	(2.6)	20	(5.1)	-2.6 (-5.4, -0.4)
Pain in extremity	91	(11.6)	49	(12.6)	-1.0 (-5.2, 2.9)
Nervous system disorders	**584**	**(74.6)**	**300**	**(77.1)**	**-2.5 (-7.6, 2.8)**
Dizziness	118	(15.1)	60	(15.4)	-0.4 (-4.9, 3.9)
Dysgeusia	128	(16.3)	49	(12.6)	3.8 (-0.6, 7.8)
Headache	234	(29.9)	113	(29.0)	0.8 (-4.8, 6.3)
Neuropathy peripheral	163	(20.8)	90	(23.1)	-2.3 (-7.5, 2.6)
Paresthesia	64	(8.2)	37	(9.5)	-1.3 (-5.1, 2.0)
Peripheral sensory neuropathy	156	(19.9)	72	(18.5)	1.4 (-3.5, 6.0)
Psychiatric disorders	**253**	**(32.3)**	**114**	**(29.3)**	**3.0 (-2.7, 8.5)**
Anxiety	61	(7.8)	31	(8.0)	-0.2 (-3.7, 2.9)
Depression	39	(5.0)	19	(4.9)	0.1 (-2.8, 2.6)
Insomnia	161	(20.6)	74	(19.0)	1.5 (-3.4, 6.2)
Renal and urinary disorders	**107**	**(13.7)**	**42**	**(10.8)**	**2.9 (-1.2, 6.6)**
Reproductive system and breast disorders	**161**	**(20.6)**	**79**	**(20.3)**	**0.3 (-4.8, 5.0)**
Breast pain	64	(8.2)	43	(11.1)	-2.9 (-6.8, 0.6)
Respiratory, thoracic, and mediastinal disorders	**407**	**(52.0)**	**184**	**(47.3)**	**4.7 (-1.4, 10.7)**
Cough	193	(24.6)	86	(22.1)	2.5 (-2.7, 7.5)
Dyspnea	99	(12.6)	50	(12.9)	-0.2 (-4.5, 3.7)
Epistaxis	117	(14.9)	63	(16.2)	-1.3 (-5.9, 3.0)
Oropharyngeal pain	66	(8.4)	38	(9.8)	-1.3 (-5.1, 2.0)
Skin and subcutaneous tissue disorders	**668**	**(85.3)**	**328**	**(84.3)**	**1.0 (-3.2, 5.6)**

Alopecia	477	(60.9)	226	(58.1)	2.8 (-3.1, 8.8)
Dermatitis acneiform	57	(7.3)	13	(3.3)	3.9 (1.2, 6.4)
Dry skin	63	(8.0)	24	(6.2)	1.9 (-1.4, 4.8)
Erythema	81	(10.3)	36	(9.3)	1.1 (-2.7, 4.5)
Nail discoloration	49	(6.3)	34	(8.7)	-2.5 (-6.0, 0.6)
Pruritus	147	(18.8)	56	(14.4)	4.4 (-0.2, 8.7)
Rash	234	(29.9)	92	(23.7)	6.2 (0.8, 11.4)
Rash maculo-papular	59	(7.5)	27	(6.9)	0.6 (-2.8, 3.6)
Vascular disorders	**248**	**(31.7)**	**121**	**(31.1)**	**0.6 (-5.2, 6.1)**
Hot flush	117	(14.9)	69	(17.7)	-2.8 (-7.5, 1.6)
Hypotension	40	(5.1)	9	(2.3)	2.8 (0.4, 4.9)

*Based on Miettinen & Nurminen method; since 95% CIs are provided without adjustment for multiplicity, they should be regarded as a helpful descriptive measure, not a formal method for assessing the statistical significance of the between-group differences in adverse experiences.

Every participant is counted a single time for each applicable row and column.

A system organ class or specific adverse event appears on this report only if its incidence in one or more of the columns meets the incidence criterion in the report title, after rounding.

Estimated differences and confidence intervals are provided in accordance with the statistical analysis plan.

Included adverse events started from the first treatment including definitive surgery and radiation therapy and up to 30 days of the last treatment including definitive surgery and radiation therapy for the non-serious adverse events and up to 90 days of the last treatment including definitive surgery and radiation therapy for the serious adverse events.

Table S7. Grade 3-5 Adverse Events Occurring at an Incidence ≥1% in One or More Treatment Groups in the Combined Phases.

Participants in Population	Pembrolizumab–Chemotherapy (N=783)		Placebo–Chemotherapy (N=389)		Difference in % vs. Placebo–Chemotherapy
	n	(%)	n	(%)	Estimate (95% CI)*
with one or more adverse events	645	(82.4)	306	(78.7)	3.7 (-1.0, 8.7)
with no adverse events	138	(17.6)	83	(21.3)	-3.7 (-8.7, 1.0)
Blood and lymphatic system disorders	**433**	**(55.3)**	**196**	**(50.4)**	**4.9 (-1.1, 11.0)**
Anemia	153	(19.5)	61	(15.7)	3.9 (-0.9, 8.3)
Febrile neutropenia	144	(18.4)	63	(16.2)	2.2 (-2.5, 6.6)
Leukopenia	33	(4.2)	16	(4.1)	0.1 (-2.6, 2.4)
Lymphopenia	7	(0.9)	4	(1.0)	-0.1 (-1.8, 1.0)
Neutropenia	276	(35.2)	134	(34.4)	0.8 (-5.1, 6.5)
Pancytopenia	14	(1.8)	5	(1.3)	0.5 (-1.3, 1.9)
Thrombocytopenia	21	(2.7)	12	(3.1)	-0.4 (-2.8, 1.5)
Cardiac disorders	**15**	**(1.9)**	**4**	**(1.0)**	**0.9 (-0.8, 2.3)**
Endocrine disorders	**25**	**(3.2)**	**0**	**(0.0)**	**3.2 (2.2, 4.7)**
Adrenal insufficiency	8	(1.0)	0	(0.0)	1.0 (0.0, 2.0)
Hypophysitis	8	(1.0)	0	(0.0)	1.0 (0.0, 2.0)
Gastrointestinal disorders	**92**	**(11.7)**	**28**	**(7.2)**	**4.6 (0.9, 7.9)**
Diarrhea	25	(3.2)	7	(1.8)	1.4 (-0.7, 3.2)
Nausea	29	(3.7)	7	(1.8)	1.9 (-0.2, 3.8)
Stomatitis	13	(1.7)	1	(0.3)	1.4 (0.1, 2.6)
Vomiting	21	(2.7)	6	(1.5)	1.1 (-0.8, 2.8)
General disorders and administration site conditions	**90**	**(11.5)**	**24**	**(6.2)**	**5.3 (1.9, 8.5)**
Asthenia	31	(4.0)	9	(2.3)	1.6 (-0.7, 3.6)
Fatigue	30	(3.8)	6	(1.5)	2.3 (0.2, 4.1)
Mucosal inflammation	8	(1.0)	3	(0.8)	0.3 (-1.3, 1.4)
Pyrexia	10	(1.3)	1	(0.3)	1.0 (-0.2, 2.1)
Hepatobiliary disorders	**24**	**(3.1)**	**2**	**(0.5)**	**2.6 (1.0, 4.1)**

Immune system disorders	**13**	**(1.7)**	**3**	**(0.8)**	**0.9 (-0.7, 2.2)**
Infections and infestations	**102**	**(13.0)**	**38**	**(9.8)**	**3.3 (-0.7, 6.9)**
Pneumonia	8	(1.0)	7	(1.8)	-0.8 (-2.7, 0.6)
Postoperative wound infection	3	(0.4)	5	(1.3)	-0.9 (-2.6, 0.1)
Sepsis	9	(1.1)	3	(0.8)	0.4 (-1.2, 1.5)
Upper respiratory tract infection	8	(1.0)	1	(0.3)	0.8 (-0.5, 1.8)
Urinary tract infection	14	(1.8)	3	(0.8)	1.0 (-0.6, 2.3)
Injury, poisoning and procedural complications	**37**	**(4.7)**	**11**	**(2.8)**	**1.9 (-0.6, 4.1)**
Infusion related reaction	9	(1.1)	2	(0.5)	0.6 (-0.8, 1.7)
Investigations	**229**	**(29.2)**	**123**	**(31.6)**	**-2.4 (-8.1, 3.1)**
Alanine aminotransferase increased	50	(6.4)	11	(2.8)	3.6 (1.0, 5.9)
Aspartate aminotransferase increased	25	(3.2)	2	(0.5)	2.7 (1.1, 4.2)
Gamma-glutamyltransferase increased	12	(1.5)	1	(0.3)	1.3 (0.0, 2.4)
Lymphocyte count decreased	12	(1.5)	5	(1.3)	0.2 (-1.6, 1.6)
Neutrophil count decreased	149	(19.0)	92	(23.7)	-4.6 (-9.8, 0.3)
Platelet count decreased	22	(2.8)	4	(1.0)	1.8 (-0.0, 3.4)
White blood cell count decreased	61	(7.8)	21	(5.4)	2.4 (-0.8, 5.2)
Metabolism and nutrition disorders	**52**	**(6.6)**	**16**	**(4.1)**	**2.5 (-0.4, 5.1)**
Hyperglycemia	5	(0.6)	6	(1.5)	-0.9 (-2.7, 0.3)
Hypokalemia	15	(1.9)	4	(1.0)	0.9 (-0.8, 2.3)
Hyponatremia	13	(1.7)	2	(0.5)	1.1 (-0.3, 2.4)
Musculoskeletal and connective tissue disorders	**20**	**(2.6)**	**4**	**(1.0)**	**1.5 (-0.2, 3.1)**
Nervous system disorders	**48**	**(6.1)**	**31**	**(8.0)**	**-1.8 (-5.3, 1.2)**
Headache	4	(0.5)	4	(1.0)	-0.5 (-2.1, 0.5)
Neuropathy peripheral	15	(1.9)	4	(1.0)	0.9 (-0.8, 2.3)
Peripheral sensory neuropathy	11	(1.4)	5	(1.3)	0.1 (-1.7, 1.5)
Polyneuropathy	2	(0.3)	4	(1.0)	-0.8 (-2.4, 0.1)
Syncope	7	(0.9)	7	(1.8)	-0.9 (-2.8, 0.4)
Psychiatric disorders	**12**	**(1.5)**	**3**	**(0.8)**	**0.8 (-0.8, 2.0)**
Renal and urinary disorders	**23**	**(2.9)**	**5**	**(1.3)**	**1.7 (-0.3, 3.3)**
Acute kidney injury	11	(1.4)	1	(0.3)	1.1 (-0.1, 2.3)

Reproductive system and breast disorders	**10**	**(1.3)**	**4**	**(1.0)**	**0.2 (-1.4, 1.5)**
Respiratory, thoracic and mediastinal disorders	**27**	**(3.4)**	**7**	**(1.8)**	**1.6 (-0.5, 3.5)**
Pulmonary embolism	15	(1.9)	2	(0.5)	1.4 (-0.1, 2.7)
Skin and subcutaneous tissue disorders	**49**	**(6.3)**	**3**	**(0.8)**	**5.5 (3.5, 7.5)**
Rash	14	(1.8)	1	(0.3)	1.5 (0.2, 2.8)
Rash maculo-papular	15	(1.9)	0	(0.0)	1.9 (0.9, 3.1)
Vascular disorders	**26**	**(3.3)**	**8**	**(2.1)**	**1.3 (-0.9, 3.1)**
Hypertension	9	(1.1)	4	(1.0)	0.1 (-1.5, 1.3)
Hypotension	9	(1.1)	1	(0.3)	0.9 (-0.4, 2.0)

*Based on Miettinen & Nurminen method; since 95% CIs are provided without adjustment for multiplicity, they should be regarded as a helpful descriptive measure, not a formal method for assessing the statistical significance of the between-group differences in adverse experiences.

Every participant is counted a single time for each applicable row and column.

A system organ class or specific adverse event appears on this report only if its incidence in one or more of the columns meets the incidence criterion in the report title, after rounding.

Estimated differences and confidence intervals are provided in accordance with the statistical analysis plan.

Included adverse events started from the first treatment including definitive surgery and radiation therapy and up to 30 days of the last treatment including definitive surgery and radiation therapy for the non-serious adverse events and up to 90 days of the last treatment including definitive surgery and radiation therapy for the serious adverse events.

Grades are based on NCI CTCAE version 4.0.

Table S8. Serious Adverse Events Occurring at an Incidence ≥1% in One or More Treatment Groups in the Combined Phases.

Participants in Population	Pembrolizumab–Chemotherapy (N=783)		Placebo–Chemotherapy (N=389)		Difference in % vs. Placebo–Chemotherapy
	n	(%)	n	(%)	Estimate (95% CI)*
with one or more adverse events	341	(43.6)	111	(28.5)	15.0 (9.2, 20.6)
with no adverse events	442	(56.4)	278	(71.5)	-15.0 (-20.6, -9.2)
Blood and lymphatic system disorders	**154**	**(19.7)**	**58**	**(14.9)**	**4.8 (0.1, 9.1)**
Anemia	20	(2.6)	9	(2.3)	0.2 (-2.0, 2.0)
Febrile neutropenia	118	(15.1)	47	(12.1)	3.0 (-1.3, 6.9)
Neutropenia	12	(1.5)	1	(0.3)	1.3 (0.0, 2.4)
Pancytopenia	11	(1.4)	4	(1.0)	0.4 (-1.3, 1.7)
Cardiac disorders	**13**	**(1.7)**	**3**	**(0.8)**	**0.9 (-0.7, 2.2)**
Endocrine disorders	**24**	**(3.1)**	**0**	**(0.0)**	**3.1 (2.1, 4.5)**
Adrenal insufficiency	8	(1.0)	0	(0.0)	1.0 (0.0, 2.0)
Hypophysitis	8	(1.0)	0	(0.0)	1.0 (0.0, 2.0)
Gastrointestinal disorders	**37**	**(4.7)**	**9**	**(2.3)**	**2.4 (0.1, 4.5)**
General disorders and administration site conditions	**42**	**(5.4)**	**9**	**(2.3)**	**3.1 (0.7, 5.2)**
Pyrexia	29	(3.7)	2	(0.5)	3.2 (1.6, 4.8)
Hepatobiliary disorders	**17**	**(2.2)**	**1**	**(0.3)**	**1.9 (0.6, 3.2)**
Immune system disorders	**11**	**(1.4)**	**1**	**(0.3)**	**1.1 (-0.1, 2.3)**
Infections and infestations	**86**	**(11.0)**	**31**	**(8.0)**	**3.0 (-0.7, 6.4)**
Pneumonia	7	(0.9)	8	(2.1)	-1.2 (-3.2, 0.2)
Postoperative wound infection	3	(0.4)	5	(1.3)	-0.9 (-2.6, 0.1)
Sepsis	9	(1.1)	4	(1.0)	0.1 (-1.5, 1.3)
Injury, poisoning and procedural complications	**23**	**(2.9)**	**4**	**(1.0)**	**1.9 (0.1, 3.5)**

Investigations	**16**	**(2.0)**	**3**	**(0.8)**	**1.3 (-0.3, 2.7)**
Metabolism and nutrition disorders	**14**	**(1.8)**	**3**	**(0.8)**	**1.0 (-0.6, 2.3)**
Nervous system disorders	**16**	**(2.0)**	**5**	**(1.3)**	**0.8 (-1.1, 2.2)**
Renal and urinary disorders	**17**	**(2.2)**	**4**	**(1.0)**	**1.1 (-0.6, 2.6)**
Acute kidney injury	8	(1.0)	1	(0.3)	0.8 (-0.5, 1.8)
Respiratory, thoracic, and mediastinal disorders	**23**	**(2.9)**	**9**	**(2.3)**	**0.6 (-1.6, 2.5)**
Pneumonitis	9	(1.1)	4	(1.0)	0.1 (-1.5, 1.3)
Pulmonary embolism	12	(1.5)	2	(0.5)	1.0 (-0.4, 2.2)
Skin and subcutaneous tissue disorders	**12**	**(1.5)**	**1**	**(0.3)**	**1.3 (0.0, 2.4)**
Vascular disorders	**11**	**(1.4)**	**3**	**(0.8)**	**0.6 (-0.9, 1.9)**

*Based on Miettinen & Nurminen method; since 95% CIs are provided without adjustment for multiplicity, they should be regarded as a helpful descriptive measure, not a formal method for assessing the statistical significance of the between-group differences in adverse experiences.

Every participant is counted a single time for each applicable row and column.

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